An Interpretable Dictionary Learning Approach for Abnormality Detection and Segmentation in Chest CT

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Abstract

Automated abnormality detection from 3D chest Computed Tomography requires clinically interpretable methods to support radiological decision-making. Current State-of-the-Art (SotA) deep learning models often rely on post-hoc explainability techniques that provide approximate localizations and lack direct traceability to the models decisions. Furthermore, SotA models rarely support simultaneous multi-class classification and grounded localization, leading to a fragmented approach to diagnosis. This also hinders the ability to benchmark models across multiple dimensions of performance, including classification accuracy, segmentation quality, and explainability. This study aims to address these challenges through two contributions: First, we introduce an inherently interpretable dictionary learning approach in which detected abnormalities can be back-projected to the original voxel space, allowing diagnostic decisions to be traced back to relevant anatomical structures. Second, we develop a unified evaluation framework to benchmark SotA models across three axes: (i) classification accuracy, (ii) segmentation quality, and (iii) explainability, using standardized metrics and a publically available 3D chest CT dataset. Our method is benchmarked against SotA models using this framework, demonstrating competitive classification and segmentation performance while providing improved interpretability compared to deep learning baselines.

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Preface

This thesis is submitted.........

Contents

List of Figures

List of Tables

Acronyms

CXR	-	Chest X-Ray
CT	-	Computed Tomography
CAD	-	Computer-Aided Diagnosis
CNN	-	Convolutional Neural Network
ViT	-	Vision Transformer
XAI	-	Explainable Artificial Intelligence
GGO	-	Ground-Glass Opacity
GGN	-	Ground-Glass Nodule
SotA	-	State of the Art

Chapter 1

Introduction

Developing a unified pipeline for multi-class abnormality classification and grounded localization addresses two of the most critical imperatives in clinical AI adoption: diagnostic accuracy and explainability. Specifically, utilizing dense pixel-level segmentation as the mechanism for localization forces the architecture to provide voxel-level visual evidence for its diagnostic claims. This ensures that the model is accountable and its claims are supported by clinical evidence.

1.1 Background

Lung disease remains a leading cause of global morbidity and mortality [1]. Research shows that early detection of pulmonary diseases significantly improves patient outcomes. For this, radiological imaging tools such as Chest X-Ray (CXR) and Computed Tomography (CT), plays an important role. However, the increasing volume of such imaging data increases radiologists' workload, leading to delays and potential errors in diagnosis. This has motivated the development of automated Computer-Aided Diagnosis (CAD) systems, which use computational algorithms and Artificial Intelligence (AI) to assist in tasks such as abnormality detection, classification, and segmentation [2].

Our research mainly focuses on the development of a CAD system using 3 Dimensional (3D) CT data. While CXR is commonly used as the first imaging test due to its low cost and low radiation exposure, studies show that their sensitivity to abnormailities is limited [3]. Comparatively, CT scans produce high resolution cross-sectional slices that helps detect smaller, early-stage, and subtle lung pathologies that are frequently missed by standard CXR.

However, CT analysis is complex and time-consuming, requiring radiologists to review hundreds of slices per scan, which are often inconsistent due to artifacts, anatomical variations and different scanners/protocols. Furthermore, different anatomical structures can share identical grayscale values. Telling them apart requires expert clinical judgment which is often subjective. High sensitivity can lead to detection of clinically insignificant abnormalities, while visual fatigue can lead to missed findings [4].

To address these challenges, recent research and clinical deployments of AI-based CAD systems are fundamentally changing how radiologists approach CT analysis. AI models such as 3D Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) have shown to reduce interpretation time, and improve lesion detection sensitivity compared to traditional manual methods. Many of these high performing models are Deep Learning based, which are criticized for their black box nature, where their internal reasoning is difficult to interpret. This lack of interpretability is a significant barrier for clinical adoption, as radiologists require evidence to support a model's diagnostic claims.

This has motivated the development of explainable AI (XAI) methods such as Gradient-weighted Class Activation Mapping (Grad-CAM) that helps to visualize exactly which parts of an image a model focuses on when making a specific prediction. Such XAI methods are highly valuable for building clinical trust in deep learning models. However, they often lack the fine-grained, pixel-level precision required to identify tiny lesions, sometimes highlighting incorrect or confounding features [5].

Recent advances in sparse representation models such as Sparse Autoencoders and Dictionary Learning have shown to provide inherent interpretability in medical imaging. This stems from their ability to reconstruct complex, high-dimensional data as a linear combination of "atoms" or features. This allows the use of efficient algorithms to easily map the output sparse domain back to the original input feature space, providing a linear relationship between the model's decision and anatomical structures. Our work builds on these advances by applying dictionary learning to 3D Chest CT data for multi-class abnormality classification and grounded localization.

1.2 Problem Statement

The clinical utility of Deep Learning based approaches is hindered by a lack of interpretability. While tools like Grad-CAM provide post-hoc explanations to model decisions, these are often insufficient for medical professionals who require a granular understanding of the model's decision making process. In high stakes environments, a model that produces a diagnosis without transparency in its reasoning is difficult to integrate into clinical workflows, as it lacks the accountability necessary for legal and ethical compliance.

1.3 Objectives and Scope

The primary goal of this study is to address the limitations of existing AI-based approaches for abnormality classification and localization in chest CTs, by developing an inherently interpretable dictionary learning framework. This approach aims to enable transparent, anatomically grounded decision making, where each classification outcome can be traced back to specific pathological patterns. This aligns with clinical reasoning and facilitates validation and adoption in radiological workflows. The specific objectives are as follows:

1.3.1 Objectives

- Implement a Dictionary Learning based CAD system using a publicly available annotated 3D chest CT dataset.
- Map dictionary atoms back to anatomical features and visualize their spatial contribution maps.

- Develop a unified evaluation framework for assessing the performance and interpretability of the proposed approach.
- Benchmark model performance against SotA classification and segmentation models on the same dataset.

1.3.2 Scope

The scope of this study focuses on the development and evaluation of a dictionary learning based approach for classification and grounded localization of lung abnormalities using volumetric 3D CT data, with the following limitations:

- This study is limited to 2 thoracic abnormalities: Pulmonary Nodules, and Ground-Glass Opacities, based on expert radiologist recommendation and dataset availbility.
- This study will utilize a single publicly available dataset for training and evaluation, which may limit generalizability to other populations and scanner types.
- The proposed approach will be limited only to interpretable methods, and will not explore approaches that provide better performance for less explainability.

Chapter 2

Literature Review

2.1 Clinical Background

Chronic respiratory diseases account for more than 4 million deaths globally per year [1]. Research shows that early detection of pulmonary diseases, particularly lung cancer and Chronic Obstructive Pulmonary Disease (COPD) dramatically lowers mortality rates. Two landmark studies, the National Lung Screening Trial (NLST) and the Nederlands-Leuvens Longkanker Screenings Onderzoek (NELSON) trial, have shown that low-dose CT (LDCT) screening reduces lung cancer mortality by 20% to 33% compared to CXR [6, 7].

2.1.1 Computed Tomography (CT)

CTs generate 3D structural information by acquiring 2D radiographic projections of an object from many angles and computationally reconstructing them into a stack of cross-sectional slices via algorithms such as filtered backprojection. Compared with other imaging modalities, CT's central advantage is its non-destructive, quantitative access to internal 3D structure. CT can visualize and quantify internal features throughout an intact volume, enabling longitudinal or repeated imaging of the same specimen over time. Attenuation-contrast CT, the dominant mode in clinical practice, works well for high-atomic-number, strongly attenuating structures such as bone or calcified tissue and, with contrast agents, vascular or soft-tissue structures. However, it provides relatively poor contrast between materials of similar composition and density (e.g., different soft tissues), a limitation that phase-contrast and other advanced modalities partially address, though these are less widely deployed clinically due to hardware and dose constraints.

Interpreting 3D chest CT specifically is complicated by several factors described in the imaging literature. Reconstructions are inherently affected by noise and artifacts, motion from breathing or cardiac movement, ring artifacts, and the partial volume effect, where a single voxel spans more than one tissue type, producing intermediate intensities that blur boundaries between structures. These effects broaden and overlap the intensity histograms of different tissues, making simple threshold-based segmentation unreliable and often necessitating boundary-based, region-growing, or learning-based segmentation methods instead. These challenges motivate the shift toward automated, learning-based segmentation approaches capable of handling such structural diversity without extensive manual tuning per dataset.

2.1.2 GGO and Pulmonary Nodules

Our study focuses on the detection and segmentation of two common pulmonary abnormalities: Ground-Glass Opacity (GGO) and pulmonary nodules. A nodule is a discrete, roughly rounded area of increased opacity within the lung parenchyma, generally classified as either a solid nodule or a ground-glass nodule (GGN) based on its internal composition. A GGO refers specifically to a hazy area of increased lung attenuation on CT that does not obscure the underlying bronchial or vascular structures passing through it. GGNs are further subdivided into pure GGNs, which contain no solid component, and mixed (subsolid) GGNs, which contain a solid portion alongside the ground-glass components.

These findings are clinically important because GGNs are frequently associated with early lung cancer and its preinvasive precursors. Notably, GGNs are not uniformly benign or malignant risk indicators equivalent to solid nodules. GGNs are, in fact, more likely to be malignant than solid nodules, and the malignant rate of mixed GGNs is higher than that of pure GGNs.

Early detection of these abnormalities is essential because lung cancer prognosis is highly stage-dependent, and most patients are still diagnosed too late for optimal outcomes. Despite improvements in treatment, the overall five-year survival rate for lung cancer remains only about 10-20% in most countries, largely because most patients are diagnosed at a late stage.

Despite their importance, GGOs and pulmonary nodules remain difficult to detect and characterize reliably. Nodules may be small relative to other anatomical structures in a chest CT, and GGO may be low-contrast against surrounding tissue. These challenges make manual interpretation difficult and motivate the shift toward automated, learning-based segmentation approaches.

2.2 Computer-Aided Diagnosis (CAD) Systems

CAD systems are computer based tools that help medical personnel in analysis of imaging studies and help in making better diagnostic decisions. CAD does not intend to replace doctors and instead, acts as a second opinion by identifying suspicious areas in medical images and providing useful information to support diagnosis. The final diagnosis is always made by the doctor[8].

2.2.1 Evolution of CAD Systems

The modern CAD framework was defined in the 1980s when the University of Chicago shifted the objective from automated diganosis to acting as a second reader [8]. Early iterations relied on manual image processing using edge detection, filtering, and feature extraction to locate nodules in chest X-rays or microcalcifications in mammograms. Chan et al. provided a pivotal study showing that radiologists using CAD identified more clustered microcalcifications than those working alone, despite a high rate of false positives [9].

Over the 1990s and 2000s, CAD spread to areas such as lung cancer detection on chest X-rays and CT scans, detection of vertebral fractures linked to osteoporosis, brain aneurysm detection using MR angiography, and tracking interval changes in bone scans [10]. Prospective studies on mammography CAD, covering tens of thousands of patients, showed a steady rise in cancer detection rates. This included finding small invasive tumors earlier. Sensitivity for microcalcification clusters rose from roughly 87% in the early 1990s to about 98% by the mid-2000s, while false positives dropped [11]. This "classical CAD" era relied on statistical pattern recognition and manually engineered features. Deep learning has since become the dominant approach in CAD.

The purpose of CAD has also grown. Early tools mostly focused on detection by flagging suspicious zones. Later versions added differential diagnosis to estimate if a nodule was malignant or benign. Some systems even retrieve similar past cases from hospital image archives (PACS) for comparison. Observer studies indicate that likelihood-of-malignancy scores improve radiologist decisions in subtle cases. Doctors still rely on their own judgment for obvious cases while CAD provides the most utility in borderline situations [**?**].

Future CAD development will likely move away from stand-alone tools. Instead, these will be integrated packages within PACS, allowing a single scan to be screened for multiple conditions simultaneously [**?**]. Deep learning has sped up this process. A single trained model can often be adapted for various abnormalities with far less manual effort than previous systems required.

In short, CAD systems evolved from early failed attempts at full automation, to assistive "second opinion" tools using hand-crafted image processing, to today's deep learning–driven systems that learn features directly from data. Deep learning has drawn more attention due to its lower need of manual feature design, better performance on complex and varied medical images and ability to scale using large datasets and modern computing power.

2.3 Deep Learning Approaches

The shift from classical CAD methods to deep learning occurred for a few reasons. Firstly, deep learning models such as CNNs and ViTs have the ability to learn specific image features from pixel data directly rather than having to manually design features and rules for all types of lesions. This makes deep learning models more flexible and often more accurate across different diseases and imaging types. Secondly, the emerging of large labeled medical imaging datasets and the availability of powerful GPU computing have made it possible to train these deep networks effectively. Thirdly, deep learning networks are more robust to image quality, patient anatomy and scanner types compared to rule based systems. Finally, they continuously outperformed classical CAD methods in pivotal studies in aspects of tumor classification and lesion detection. The subsequent sections review some of the popular deep learning models used in medical imaging research, and also used in this study.

2.3.1 nnU-Net

The nnU-Net addresses a persistent problem in biomedical image segmentation. Even well-established architectures like the U-Net require extensive, dataset-specific manual tuning of preprocessing, network topology, and training parameters. Small misconfigurations can cause large performance drops. Rather than proposing a new architecture, nnU-Net systematizes this configuration process itself. It decomposes design decisions into three categories:

1. Fixed parameters that generalize across datasets (e.g., architecture template, loss function, optimizer).

2. Rule-based parameters derived automatically from a "dataset fingerprint" (image size, spacing, intensity distribution, modality) via heuristics (e.g., target spacing, patch size, batch size, network depth).

3. A small set of empirical parameters selected via cross-validation (choice among 2D, 3D, and 3D cascade configurations, and whether to apply post-processing such as largest-component suppression).

Figure 2.1 shows the nnU-Net configuration method:

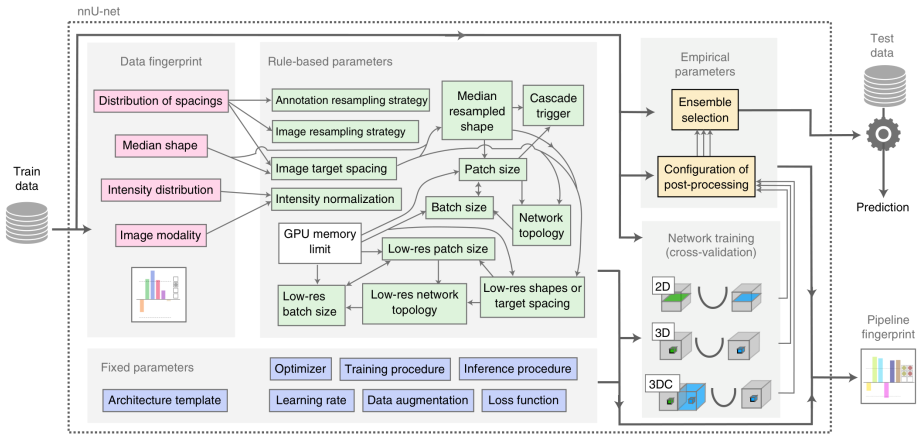

Figure 2.1: nnU-Net configuration method

Evaluated across 23 public datasets and 53 segmentation tasks spanning MRI, CT, electron microscopy, and fluorescence microscopy, nnU-Net outperformed most specialized, task-tuned pipelines without any manual intervention, setting a new state of the art on 33 of 53 target structures. Notably, the authors' analysis of the KiTS 2019 leaderboard showed that configuration choices such as patch size, spacing, normalization affected performance more than architectural modifications. Figure 2.2 shows example segmentations produced by nnU-Net on different datasets:

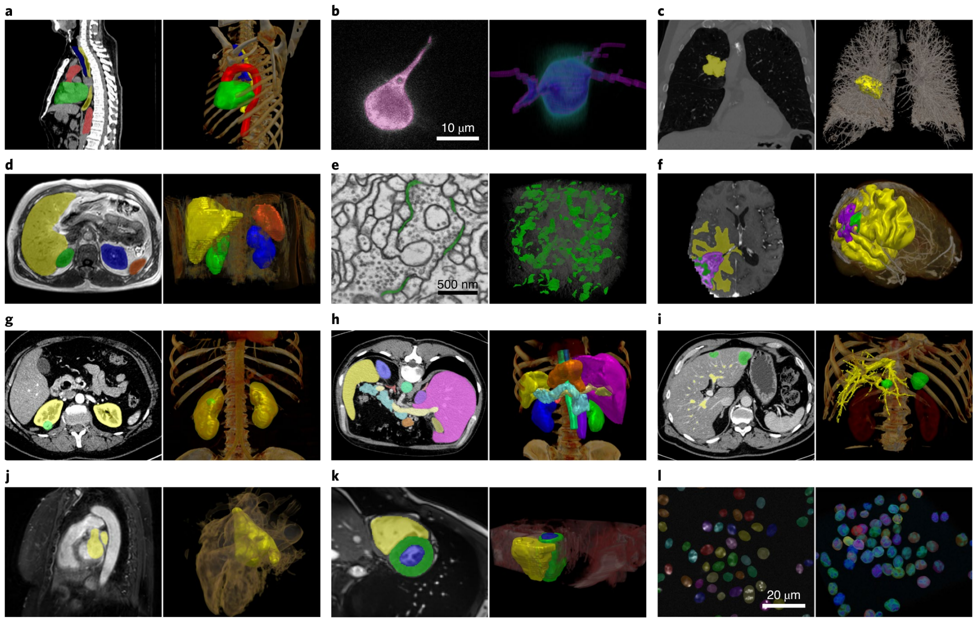

Figure 2.2: Example segmentations by nnU-Net

This suggests that careful pipeline configuration is often the primary bottleneck in medical image segmentation. This positions nnU-Net less as a competing architecture and more as a strong, reproducible baseline and out-of-the-box framework, and it is frequently cited in the literature as the standard benchmark against which new segmentation methods are compared.

2.3.2 BioMedParse

BiomedParse is a "Foundation Model for Joint Segmentation, Detection, and Recognition of Biomedical Objects Across Nine Modalities" [12]. Rather than relying on a fixed label set during inference, the model accepts a natural language prompt describing the target anatomy or pathology and produces a pixel level mask for the queried concept. This makes BiomedParse especially useful in settings where the structures of interest are heterogeneous, difficult to enumerate in advance, or described differently across institutions and reports.

The model combines image encoding with language understanding and uses a transformer based decoder to align the prompt with the visual features extracted from the scan [12]. Figure 2.3 shows the flowchart of the architecture:

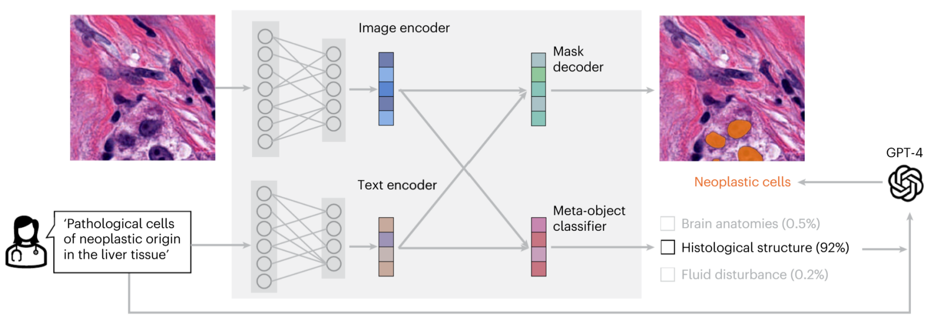

Figure 2.3: BiomedParse architecture

The output is a segmentation mask, as shown in Figure 2.4.

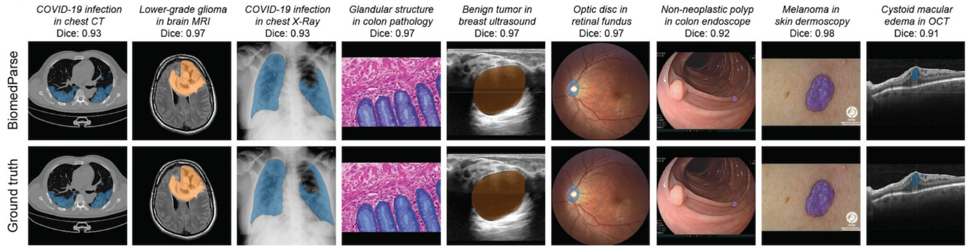

Figure 2.4: Example outputs of BiomedParse

BiomedParse serves as a SotA reference model for discussing localization quality, and interpretability in our study.

2.3.3 TotalSegmentator

Many deep learning methods were developed to segment only one or a few organs. Wasserthal et al. introduced TotalSegmentator, a deep learning framework that can automatically segment 104 anatomical structures from a single CT scan [13]. Figure 2.5 shows the main classes of TotalSegmentator, which include 27 organs, 59 bones, 10 muscles, 8 vessels.

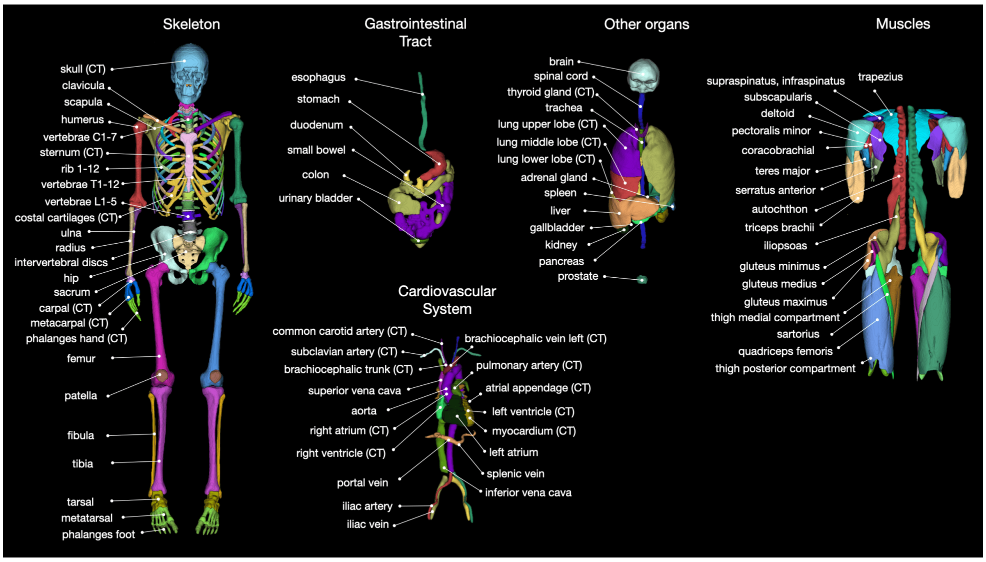

Figure 2.5: Main classes of TotalSegmentator

The framework is built on the nnU-Net architecture and was trained using more than 1,200 routine clinical CT scans collected from different hospitals and patient groups [13]. It achieved a mean Dice Similarity Coefficient (DSC) of about 0.94. Because of its high segmentation accuracy, TotalSegmentator is commonly used as a preprocessing tool in medical image research [14].

2.3.4 Merlin

Merlin is an AI model that can help interpret CT scans automatically. 3D vision-language model (VLM) built to understand abdominal CT scans by learning jointly from the images, radiology reports, and electronic health record (EHR) data [15].

Most earlier medical vision-language models were built for 2D images, such as chest X-rays, paired with short captions. CT scans are different, they are 3D volumes made of hundreds of slices, and simply applying a 2D model slice-by-slice throws away important 3D spatial information, like how a lesion connects across neighboring slices. Merlin was built to fix this by processing the full 3D CT volume directly, instead of treating it as a stack of unrelated 2D images [15].

Merlin is trained using over 6 million CT images from more than 15,000 scans, paired with over 1.8 million EHR diagnosis codes and more than 6 million tokens of radiology report text [15]. Instead of relying only on manually labeled data, which is slow and expensive to collect, Merlin learns from data hospitals already generate: doctors' reports and diagnosis codes. Merlin follows the concept introduced by CLIP, a foundational vision-language model designed originally for natural images. In CLIP style training, similar or matching image or text pairs closer to each other in a shared representation space while non-matching pairs are pushed apart making the model learn which images and text describe the same thing. Merlin adapts this idea to 3D CT data and adds EHR code supervision on top, which earlier medical VLMs generally did not use [15].

Merlin is part of a broader wave of work bringing vision-language modeling into 3D medical imaging, alongside similar efforts like CT-CLIP, which uses the same contrastive approach but focuses on chest CT scans[15]. Overall, Merlin shows that learning directly from full 3D volumes and everyday hospital data, rather than small 2D datasets with manual labels, can produce useful for reducing radiologist workload.

2.3.5 MedSAM

MedSAM adapts the Segment Anything Model (SAM) developed by Meta AI, originally developed for natural images, into a general-purpose segmentation model for medical scans [16]. Rather than training a separate network for each organ or modality, a single fine-tuned model is reused across many segmentation tasks. A key limitation is that MedSAM operates on 2D slices. Applying it to a 3D chest CT volume therefore requires a separate bounding box on every slice, and the resulting masks do not capture how a nodule connects across neighbouring slices.

MedSAM2 addresses this limitation by building on the SAM2 architecture and its memory-attention mechanism, allowing a single prompt to be propagated across an entire volume instead of being redrawn slice by slice, which substantially reduces the annotation effort required [17].

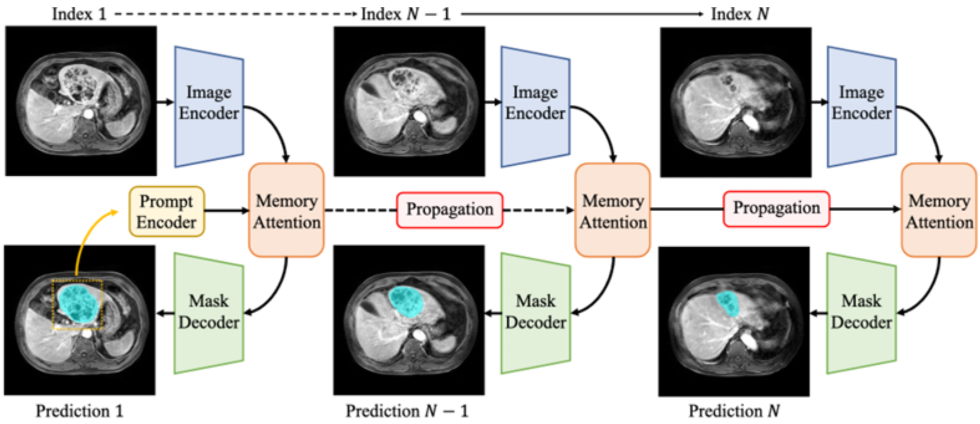

Figure 2.6: Architecture of MedSAM2

Despite this improvement, pulmonary findings remain difficult for these models to segment accurately. A benchmarking study comparing MedSAM and MedSAM2 against standard segmentation networks found that both achieved strong performance on whole-lung segmentation but performed noticeably worse on individual tumours and small nodules, with results sensitive to the quality and consistency of the box prompts [18]. This reflects a broader limitation of prompt-driven segmentation models.

2.3.6 CT-CLIP

CT images contain valuable information for diagnosing pulmonary diseases and are used by many Deep Learning approaches as the only source of information. However, radiology reports provide detailed clinical descriptions. Hamamci et al. proposed Computed Tomography Contrastive Language-Image Pre-training (CT-CLIP) [19], a vision-language foundation model for chest CT images, which utilizes both modalities. Unlike traditional supervised models that require large amounts of labeled data, CT-CLIP learns the relationship between CT scans and their corresponding radiology reports using contrastive learning. This allows the model to learn meaningful image and text representations without task-specific annotations.

CT-CLIP consists of a 3D image encoder and a text encoder. The image encoder extracts features from CT volumes, while the text encoder processes the corresponding radiology reports. Figure 2.7 shows the architecture of CT-CLIP.

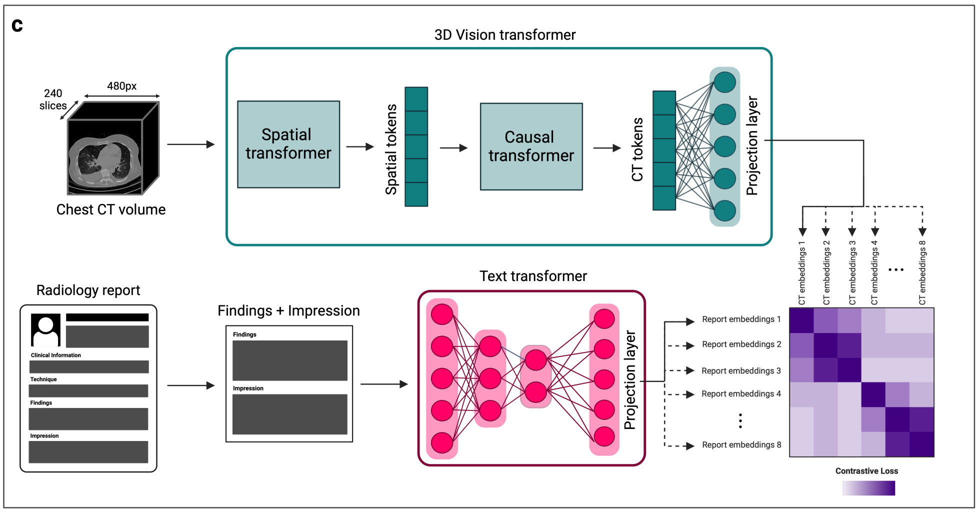

Figure 2.7: CT-CLIP architecture

During training, the model learns to match each CT scan with its correct report in a shared feature space. The model was trained using the CT-RATE dataset [20], which pairs over 50,000 reconstructed 3D chest CT volumes from around 25,700 scans of over 21,000 patients with their corresponding radiology reports. It has shown strong performance in pulmonary disease classification, image retrieval, and zero-shot learning tasks, outperforming fully supervised models across key metrics [19].

2.4 Explainable AI (XAI) in Medical Imaging

Artificial intelligence (AI) has become an important tool in medical imaging by supporting tasks such as disease detection, classification, and image analysis. Many deep learning models achieve high accuracy, but their decision-making process is often difficult to understand. These models are commonly referred to as "black-box" models because they do not clearly explain how a prediction is made [21, 22]. In healthcare, this lack of transparency can reduce clinicians' confidence in AI-assisted diagnosis.

Interpretability is important because medical decisions directly affect patient care. Clinicians need to understand why a model predicts a particular disease before using its results in clinical practice. Explainable AI (XAI) provides methods that help users understand how AI models make decisions. This improves trust, supports clinical decision-making, and helps identify possible model errors or biases [23].

The importance of XAI has also increased due to ethical and regulatory requirements. Healthcare AI systems are expected to be transparent, reliable, and accountable before they are used in real clinical environments. Explainable models help meet these requirements by providing evidence that supports their predictions instead of producing results without explanation [22, 23].

Explainability is especially important in medical imaging because subtle abnormalities can be difficult for even experienced clinicians to identify consistently. When an AI system flags a region of a scan as diagnostically significant, clinicians need to know which features or regions influenced that prediction. This helps verify the diagnosis, exposes potential model errors or biases before they affect patient care, and increases confidence in AI-assisted clinical decisions. Therefore, improving interpretability has become an important research direction in computer-aided diagnosis systems [21, 23].

2.4.1 SHAP

SHapley Additive exPlanations (SHAP) is a widely used explainable artificial intelligence (XAI) method for interpreting machine learning models. It is based on Shapley values from cooperative game theory, where each feature is assigned an importance value according to its contribution to the final prediction. This provides a consistent and theoretically sound approach for explaining both traditional machine learning and deep learning models [24].

SHAP provides local explanations by design: for a given prediction, it attributes a signed contribution to each input feature, showing whether that feature increased or decreased the predicted outcome relative to a baseline expectation [24]. Building on this local formulation, later work showed that many local SHAP explanations can be aggregated across a dataset to reveal global patterns in a model's behavior, such as which features matter most overall or how risk factors interact across a population [25]. Together, these properties help users understand why a model makes a particular decision and improve confidence in AI-based systems [23].

In medical imaging, SHAP has been used to improve the transparency and interpretability of artificial intelligence models. It helps researchers and clinicians identify the imaging features and clinical variables that have the greatest influence on model predictions. This makes it easier to determine whether the model is learning meaningful medical information rather than irrelevant patterns. As a result, SHAP supports the development of more reliable and trustworthy computer-aided diagnosis systems [23].

For pulmonary disease analysis using CT images, SHAP has been applied to highlight the image regions and features driving a model's predictions. For example, in CNN-based lung cancer classification from CT scans, SHAP attributions have been shown to emphasize irregular, dense tissue and spiculated margins for malignant predictions, while benign predictions were associated with smooth, well-defined nodule boundaries [26]. These explanations help clinicians verify whether a model focuses on clinically relevant information and identify possible prediction errors. Therefore, SHAP improves the interpretability, reliability, and clinical acceptance of deep learning models used for pulmonary disease diagnosis.

2.4.2 LIME

Local Interpretable Model-agnostic Explanations (LIME) is a popular XAI method used to explain the predictions of machine learning models. It is model-agnostic, which means it can be used with any type of model, no matter how it works internally. LIME explains one prediction at a time by building a simple model that approximates how the original model behaves near that prediction [27].

For image data, LIME first divides the image into small regions, called superpixels. In medical images, these regions can also be meaningful patches of the scan. LIME then creates many versions of the image by hiding or changing some of these regions and checks how the model's prediction changes each time. A simple linear model is then fitted to these results, and the regions with the highest importance are shown as the explanation [23, 27].

In medical imaging, this helps researchers and doctors see which parts of an image the model used to make its decision. This makes it easier to check if the model is looking at real medical features instead of irrelevant patterns [23, 28]. For pulmonary disease detection using CT scans, LIME can point out the regions linked to abnormalities, helping clinicians confirm that the model is making sensible decisions and catch possible errors [23].

However, LIME also has some drawbacks. Its explanations can change between runs because of the randomness in how it perturbs the input, and the results depend on settings like the number of samples used [27]. It is also slow to run on large 3D CT scans, since each perturbed sample needs a separate prediction from the model. Because of this, gradient-based methods like Grad-CAM are often preferred for 3D medical imaging tasks [23].

2.4.3 Grad-CAM

Gradient-weighted Class Activation Mapping (Grad-CAM) is a popular XAI method used to explain the predictions of convolutional neural networks (CNNs). It is not model-agnostic like LIME. It only works with CNN models, because it uses the gradients from the last convolutional layer to find out which parts of the image the model looked at [29].

Grad-CAM works by looking at the feature maps from the last convolutional layer. It calculates the gradient of the predicted class score with respect to these feature maps. These gradients are averaged to get an importance weight for each feature map. The feature maps are then combined using these weights and passed through a ReLU function, so only the parts with a positive effect on the prediction are kept. This gives a coarse heatmap that shows the regions of the image that mattered most for the prediction [29].

In medical imaging, Grad-CAM is used a lot because it is fast and can be applied to any CNN without changing the model. It helps check if a deep learning model is looking at real abnormalities, like tumours or lesions, instead of random background areas [5, 23]. For pulmonary disease detection using CT images, Grad-CAM can highlight areas linked to nodules or ground-glass opacities (GGOs), helping doctors confirm the model's decision is based on meaningful areas [23].

However, Grad-CAM has some limitations. Since it uses the last convolutional layer, its heatmap has lower resolution than the input image and needs to be upsampled. This can make the explanation blurry and less accurate for small abnormalities [29]. It can also struggle when there are multiple objects of the same class in an image, since it may highlight them all together instead of separately [5].

2.4.4 Grad-CAM++

Grad-CAM++ is an improved version of Grad-CAM. It was made to fix some of Grad-CAM's problems, especially when an image has more than one object of the same class, or when the object is small [30].

The main difference is how the importance weight for each feature map is calculated. Grad-CAM uses a simple average of the gradients. Grad-CAM++ uses a weighted combination of higher-order derivatives instead. This gives more importance to the pixels that contribute the most to the prediction, instead of treating all pixels equally [30]. Because of this, Grad-CAM++ can locate objects better when there are multiple instances of the same class, and it can highlight the full object more accurately than Grad-CAM.

In medical imaging, this is useful when there is more than one abnormality of the same type in a scan, like multiple lung nodules in one CT slice. Grad-CAM++ can separate and highlight each nodule clearly, giving a fuller explanation than Grad-CAM, which may only focus on the biggest one [5, 30]. For pulmonary CT analysis, this means Grad-CAM++ can help doctors see all the important abnormal areas in a scan, not just the most obvious one. This matters because missing a small nodule could lead to a missed diagnosis.

However, Grad-CAM++ also has drawbacks. It costs more to compute than Grad-CAM, since it needs to calculate higher-order gradients [30]. It also has the same resolution problem as Grad-CAM, since it also relies on the last convolutional layer and needs upsampling to match the input image size [5].

2.4.5 Other CAM Methods

Besides Grad-CAM and Grad-CAM++, several other CAM-based methods have been proposed to address specific weaknesses in the original approach. Three common ones are HiResCAM, Score-CAM, and XGrad-CAM.

HiResCAM was introduced to fix a problem with how Grad-CAM averages gradients. This averaging step can sometimes make Grad-CAM highlight regions that the model did not actually use for its prediction. HiResCAM solves this by multiplying the gradients with the feature maps directly, instead of averaging the gradients first. This produces an explanation that is more faithful to what the model actually used to make its decision [31]. HiResCAM has been applied to 3D medical imaging tasks, such as detecting abnormalities across CT slices, where faithful localisation is especially important [31].

Score-CAM takes a different approach. Instead of using gradients at all, it measures the importance of each feature map by masking the input image with that feature map and checking how much the model's confidence score changes. Feature maps that cause a bigger increase in confidence are given a higher weight [32]. Because it does not use gradients, Score-CAM avoids some of the noise and instability that gradient-based methods can have. However, it is much slower, since it needs a separate forward pass through the model for every feature map [32].

XGrad-CAM was designed to make Grad-CAM more mathematically consistent. The authors introduced two properties, called sensitivity and conservation, that a good explanation method should satisfy. Grad-CAM does not fully satisfy these properties, so XGrad-CAM changes how the importance weights are calculated to better meet them [33]. In practice, this gives XGrad-CAM better localisation than Grad-CAM while still being fast to compute, since it does not need the extra higher-order gradients that Grad-CAM++ requires [33].

Each of these methods offers a different trade-off. HiResCAM improves faithfulness with almost no extra cost, Score-CAM improves stability at the cost of speed, and XGrad-CAM improves theoretical grounding while staying efficient. For 3D medical imaging tasks such as pulmonary CT analysis, this means the choice of method should depend on whether faithfulness, stability, or speed matters most for the task at hand [31–33].

2.4.6 Limitations of XAI Methods

XAI methods like LIME and Grad-CAM have made deep learning models easier to understand. But they still have some important limitations.

One issue is that many of these methods only give an approximate explanation, not an exact one. LIME builds a simple model around a prediction to explain it. But this simple model may not always match the real model well, especially for complex, non-linear networks [22, 27].

Another problem is instability. Running the same method on the same input more than once can give slightly different results. This makes it harder for doctors to trust the explanation if it keeps changing [23, 27].

Gradient-based methods like Grad-CAM have their own issues too. They rely on the last convolutional layer, so the heatmap has low resolution and needs to be upsampled. This can make it harder to localise small abnormalities precisely [5, 29].

Most XAI methods also only show where the model looked, not why it made that decision. They do not explain the deeper reasoning behind a prediction [21, 34]. This is a real problem in medical imaging, since doctors often need to understand the reasoning, not just the location.

Finally, there is no single standard way to measure how good an explanation is. Different XAI methods are often evaluated with different metrics, which makes it hard to compare them fairly [22, 23].

Because of these limitations, XAI methods should be used carefully. Their explanations need to be checked before being trusted in real clinical decisions.

2.5 Sparse Representation

Why is sparsity useful? How is sparse representation learned?

2.5.1 Sparse Coding

Sparse coding is a signal representation framework in which a signal is expressed as a linear combination of only a small number of elements ("atoms") drawn from a dictionary, rather than as a dense combination over an orthogonal basis. Formally, given a signal x and a dictionary D whose columns are atoms, sparse coding seeks a coefficient vector α such that:

$$x \approx D\alpha,$$

with α containing mostly zero entries. A key aspect of this framework is the use of overcomplete representations, where the dictionary D has more atoms than the dimensionality of the signal itself (i.e., more columns than rows). This redundancy is deliberate: it gives the representation greater flexibility to match the underlying structure of the signal, allowing different signals to be well-approximated by different small subsets of atoms, which improves representational power, robustness to noise, and the ability to capture diverse structural patterns compared to a complete (non-redundant) basis. The trade-off is that recovering α becomes an underdetermined and generally NP-hard combinatorial problem, since many possible coefficient vectors could reconstruct x, and one must impose sparsity as a constraint or penalty to select a meaningful solution.

Because exact sparsest-solution search is computationally intractable, two main families of practical algorithms are used:

1. Greedy Algorithms (Matching Pursuit, OMP)
2. Convex relaxation Algorithms (LARS, FISTA)

Greedy methods iteratively select the atom most correlated with the current residual, add it to the active support set, and update the coefficients, repeating until a stopping criterion (sparsity level or residual error) is met; these methods are fast and simple but only approximate the optimal solution.

Convex relaxation methods instead replace the intractable $\ell 0$-norm (which counts nonzero entries) with the $\ell 1$-norm, yielding a convex optimization problem—as in Basis Pursuit or the Lasso—that can be solved reliably and, under certain conditions on the dictionary (e.g., restricted isometry or incoherence properties), provably recovers the same sparse solution as the original $\ell 0$ problem. In practice, greedy methods are often preferred for speed and low-sparsity regimes, while convex relaxation approaches tend to offer stronger theoretical guarantees and more stable solutions in noisier or higher-sparsity settings.

2.5.2 Matching Pursuit (MP)

Matching Pursuit is a greedy algorithm for approximating a signal x as a sparse combination of atoms from a dictionary D. It proceeds iteratively: at each step, it computes the inner product between the current residual (initially the signal itself) and every atom in the dictionary, selects the atom with the highest absolute correlation, and updates the coefficient for that atom by projecting the residual onto it. The residual is then updated by subtracting this contribution, and the process repeats until a stopping criterion is reached (target sparsity or residual energy threshold). MP is simple and computationally cheap, but because it only updates one coefficient per iteration without adjusting previously selected coefficients, it can select the same atom multiple times and converges slowly, and its solutions are generally suboptimal compared to more refined methods.

2.5.3 Orthogonal Matching Pursuit (OMP)

OMP improves on MP by re-optimizing all currently selected coefficients at every iteration rather than only the most recently added one. At each step, it selects the atom most correlated with the residual (as in MP), adds it to the active support set, and then solves a least-squares problem over all atoms currently in the support to obtain new coefficients that minimize the residual jointly. Because the residual is then orthogonal to all previously selected atoms, OMP never reselects the same atom twice, and it typically converges in fewer iterations and yields more accurate reconstructions than plain MP. The added cost is the least-squares solve at each iteration, making OMP more computationally expensive per step, though still tractable for moderate sparsity levels.

2.5.4 Least Angle Regression (LARS)

LARS is an efficient algorithm for computing the entire regularization path of the Lasso (ℓ1-penalized least squares) problem. Rather than fixing a penalty parameter λ and solving a single optimization, LARS starts with all coefficients at zero and iteratively identifies the predictor (atom) most correlated with the current residual, moving the coefficient in that predictor's direction. Instead of moving all the way to the least-squares solution as in stepwise regression, LARS moves only until another predictor becomes equally correlated with the residual, at which point that predictor is added to the active set and the direction is updated to remain equiangular between all active predictors. With a simple modification (allowing coefficients to be dropped from the active set when they cross zero), LARS exactly reproduces the full Lasso solution path with a computational cost comparable to ordinary least squares, making it a popular and efficient tool for convex-relaxation-based sparse coding, particularly when the full path of solutions across sparsity levels is of interest.

2.5.5 Fast Iterative Shrinkage-Thresholding Algorithm (FISTA)

FISTA solves ℓ1-regularized sparse coding problems, where the goal is to minimize:

$$\sqrt{\|x - D\alpha\|}^2 + \lambda\|\alpha\|_1$$

It builds on the Iterative Shrinkage-Thresholding Algorithm (ISTA), which alternates between a gradient descent step on the data-fidelity term and a soft-thresholding (shrinkage) step that enforces sparsity by shrinking small coefficients toward zero. While ISTA converges at a sublinear rate of O(1/k), FISTA accelerates this by incorporating a momentum term—extrapolating using a weighted combination of the current and previous iterates—which improves the convergence rate to O(1/k$^{\mathbf{2}}$) without increasing per-iteration cost. This makes FISTA well-suited to large-scale sparse coding and compressed sensing problems where fast, reliable convergence to the ℓ1-regularized solution is needed.

2.6 Dictionary Learning

Dictionary learning is a technique in which an image patch or a region of a medical scan is represented as a combination of a small number of basic building blocks, rather than by describing every pixel value directly. This idea comes from sparse coding theory and is widely used in image denoising, compression, and interpretable machine learning, including medical image analysis [35] [36].

A dictionary is a matrix, written as $D \in \mathbb{R}^{n\times k}$, where each column d_i is called an atom. Each atom is a small, fixed pattern, for example a short edge, a texture, or a simple shape, that acts like a basic building block for describing images [35].

$$D = [\, d_1, d_2, \ldots, d_k \,], \qquad \|d_i\|_2 = 1 \;\forall i$$

Each atom is normalized to unit length so that none of them dominate because of scale. A good dictionary has atoms that are simple and reusable.

Once a dictionary is available, any image $x \in \mathbb{R}^n$ is approximated as a weighted sum of a small number of atoms:

$$x \approx D\alpha = \sum_{i=1}^{k} \alpha_i d_i$$

Here, $\alpha \in \mathbb{R}^k$ is the sparse code; most of its values are zero, and only a few atoms are actually used to rebuild the image. Finding this code is called sparse coding, and it can be solved by minimizing the reconstruction error together with a sparsity penalty [36] [37]:

$$\min_{\alpha} \ \frac{1}{2}\|x - D\alpha\|_2^2 + \lambda\|\alpha\|_1$$

This is the Lasso formulation, where λ controls how sparse the solution is [38]. When the dictionary itself is learned from training data $X = [x_1, \ldots, x_m]$, both D and the codes $A = [\alpha_1, \ldots, \alpha_m]$ are optimized together

$$\min_{D,A} \ \|X - DA\|_F^2 + \lambda\|A\|_1 \quad \text{subject to} \quad \|d_i\|_2 = 1$$

Learned atoms are not random they turn out to look like meaningful patterns. Scince each image patch is reconstructed using only a small number of atoms. Each atoms tends to specialize in representing a simple pattern rather than blending many patterns together. This lets a person actually look at a lerned dictionary and understand what each atom represents, wich is must harder with internal weights of a deep neural network. In medical imaging, atoms can end up representing specific tissue textures or lesion patterns that a radiologist can vissually recognize[39].

Learning a dictionary is usually done by fixing the dictionary and solving for sparse codes, then fixing the codes and updating the dictionary. The sparse coding step is commonly solved with Orthogonal Matching Pursuit (OMP), a greedy method that adds one atom at a time [40]

$$i^* = \arg\max_{i} \ |d_i^T r|, \qquad r = x - D\alpha$$

or with Lasso/LARS, using the ℓ_1-penalized objective shown above [38]. The dictionary update step is where the main algorithms differ, covered below.

2.6.1 K-Means Singular Value Decomposition (K-SVD)

K-SVD is one of the most widely used algorithms for learning a dictionary, and it solves the following optimization problem[38]

$$\min_{D,X} \left\{\|Y - DX\|_F^2\right\} \quad \text{subject to} \quad \forall i, \ \|x_i\|_0 \leq T_0$$

Here, Y is the matrix of training data, for example CT image patches, D is the dictionary being learned, and X is the matrix of sparse codes, where each column x_i is the code for one training sample. The constraint $\|x_i\|_0 \leq T_0$ means that each code is allowed to use at most T_0 atoms, using the ℓ_0 norm to directly count non-zero entries rather than approximating sparsity with an ℓ_1 penalty like Lasso does. This gives K-SVD tighter control over exactly how sparse each representation is.

K-SVD solves this problem by alternating between two steps. First, with the dictionary D fixed, it finds the sparse codes X using a pursuit algorithm such as Orthogonal Matching Pursuit. Second, with the codes fixed, it updates the dictionary one atom at a time instead of all at once, which makes it converge faster and produce sharper, more specific atoms [38].

2.6.2 Method of Optimized Directions (MOD)

Method of Optimized Directions is an earlier and simpler approach than K-SVD. Instead of updating atoms one at a time, MOD updates the entire dictionary at once, using a direct least-squares solution [?]. Given the training data X and current sparse codes A, the optimal dictionary is:

$$D = XA^T(AA^T)^{-1}$$

This comes directly from setting the derivative of the reconstruction error $\|X - DA\|_F^2$ to zero and solving for D. MOD is fast and easy to implement, but because it updates all atoms together, it tends to need more iterations to converge compared to K-SVD, and can get stuck producing less sharp atoms [?].

2.6.3 Online Dictionary Learning (ODL)

Both K-SVD and MOD assume the entire training dataset fits in memory and is processed all at once, which becomes impractical for very large image datasets. Online Dictionary Learning solves this by processing the data in small batches, updating the dictionary incrementally as new samples arrive, instead of waiting to see the whole dataset [?]. At each step t, a new sample x_t is drawn, its sparse code α_t is computed using the current dictionary, and then the dictionary is updated using accumulated statistics from all samples seen so far:

$$A_t = A_{t-1} + \alpha_t \alpha_t^T, \qquad B_t = B_{t-1} + x_t \alpha_t^T$$

$$D_t = \arg\min_D \frac{1}{t} \sum_{i=1}^{t} \left(\frac{1}{2} \|x_i - D\alpha_i\|_2^2 \right)$$

which can be solved efficiently using A_t and B_t instead of storing all past samples. This makes ODL well suited for large-scale problems, including learning dictionaries from thousands of medical image patches, since memory use stays constant regardless of dataset size [?].

2.7 Discriminative Dictionary Learning

Traditional dictionary learning methods mainly focus on minimizing the reconstruction error between the input data and its sparse representation [41]. However, good reconstruction does not necessarily produce features that are useful for classification. Samples from different classes may activate similar atoms if the learned dictionary only captures common structures in the data.

Discriminative dictionary learning addresses this limitation by incorporating class information during dictionary optimization. The objective is not only to represent the input samples accurately but also to learn sparse codes that improve the separation between different classes. By learning class aware atoms, discriminative dictionary learning can provide both classification capability and a more interpretable representation. Several approaches have been proposed, including Label Consistent K-SVD (LC-KSVD) and Fisher Discriminative Dictionary Learning (FDDL), which extend the K-SVD framework by introducing supervised learning constraints.

2.7.1 Label Consistent K-SVD (LC-KSVD)

Label Consistent K-SVD (LC-KSVD), introduced by Jiang et al. [42], extends the classic K-SVD framework by embedding label consistency into the dictionary update process. The core idea is to encourage sparse codes from samples of the same class to exhibit similar activation patterns, while those from different classes diverge. This is achieved without sacrificing the algorithm's efficient iterative structure involving sparse coding and dictionary refinement.

The method comes in two main variants. LC-KSVD1 incorporates a label consistency term that aligns the sparse codes with an ideal discriminative structure through a linear transformation:

$$\min_{D,A,A_c} \|X - DA\|_F^2 + \alpha\|Q - A_c A\|_F^2 \quad \text{s.t.} \quad \|a_i\|_0 \leq T \ \forall i. \tag{2.1}$$

Here, Q is a target matrix encoding class specific indicator patterns, and A_c learns to map codes toward this structure. LC-KSVD2 extends LC-KSVD1 by adding a classifier learning term. The dictionary, sparse codes, label consistency transformation, and classifier are jointly optimized:

$$\min_{D,A,A_c,W} \|X - DA\|_F^2 + \alpha\|Q - A_c A\|_F^2 + \beta\|H - WA\|_F^2 \quad \text{s.t.} \quad \|a_i\|_0 \leq T, \tag{2.2}$$

where H contains the label information. The additional classification term allows the learned sparse representation to be directly optimized for prediction. Due to its ability to learn class specific atoms, LC-KSVD has been applied in image classification tasks where interpretability of learned features is important [42].

2.7.2 Fisher Discriminative Dictionary Learning (FDDL)

Fisher Discriminative Dictionary Learning (FDDL), proposed by Yang et al. [43], is another supervised dictionary learning approach that directly enforces Fisher like criteria on both the dictionary atoms and the sparse coefficients. It learns a structured dictionary $D = [D_1, D_2, \ldots, D_c]$ with class specific sub dictionaries D_i while promoting sparse codes A that exhibit low within class scatter and high between class scatter.

The core objective combines a discriminative fidelity term $r(\cdot)$ (ensuring good reconstruction within class and poor reconstruction across classes) with a Fisher discrimination penalty on coefficients:

$$\min_{D,X} \sum_i r(A_i, D, X_i) + \lambda_1\|X\|_1 + \lambda_2\Big(\operatorname{tr}(S_W(X)) - \operatorname{tr}(S_B(X)) + \eta\|X\|_F^2\Big), \tag{2.3}$$

where $r(\cdot)$ encourages each class to be well represented by its own sub dictionary while poorly represented by others. S_W and S_B denote the within class and between class scatter matrices of the sparse codes. The elastic term controlled by η is added to guarantee that the overall Fisher discrimination function remains convex, which greatly improves optimization stability [43].

Compared with LC-KSVD, which introduces label consistency through a learned transformation and classifier, FDDL focuses on improving the statistical distribution of sparse codes. This makes it suitable for classification tasks where the separation between classes is important.

2.7.3 Frozen Dictionary Learning

While most discriminative methods update the entire dictionary during training, certain scenarios benefit from preserving previously acquired knowledge. Frozen dictionary learning, as introduced in the Outlier Learning via Augmented Frozen (OLAF) Dictionaries framework [44], addresses this by fixing a subset of atoms learned from baseline (normal) data while allowing new atoms to adapt to anomalous samples.

The composite dictionary takes the partitioned form $D = [D_f \mid D_u]$, where D_f remains frozen after initial training on normal data, and only the unfrozen portion D_u is updated on data containing deviations. Both K-SVD and Alternating Minimization algorithms can be modified straightforwardly to respect this constraint by skipping updates on frozen columns during the dictionary refinement stage [44].

This approach is particularly useful when preserving existing knowledge is important. In medical imaging, a dictionary learned from normal anatomy can be kept fixed, while additional atoms are learned from abnormal cases. This separation helps maintain normal structural information and improves the interpretability of disease related atoms. However, the effectiveness of this method depends on the quality of the initial frozen dictionary, since an incomplete initial representation may limit the ability of the model to adapt to new patterns.

2.7.4 Dictionary Learning in Medical Imaging

Dictionary learning has been applied to many areas of medical image analysis over the past two decades. The main idea is to learn a set of atoms directly from the data, so that each image or patch can be represented as a sparse combination of these atoms. This approach was first popularised by the K-SVD algorithm, which learns an overcomplete dictionary by alternating between sparse coding and dictionary updateq steps [41]. Later, Jiang et al. extended this idea to classification tasks by proposing Label Consistent K-SVD (LC-KSVD), which adds a label consistency term so that the learned dictionary is more discriminative for classification [42].

In medical imaging, dictionary learning has been used for a wide range of tasks. Ravishankar and Bresler used dictionary learning to reconstruct MRI images from undersampled k-space data, allowing shorter scan times without losing image quality [45]. Tong et al. applied discriminative dictionary learning to segment the hippocampus from brain MRI scans, showing that sparse coding can capture useful anatomical patterns for segmentation [46]. In CT imaging, Ben-Cohen et al. combined a fully convolutional network with sparsity-based dictionary learning to detect liver lesions, showing that dictionary learning can also be used alongside deep learning models rather than as a replacement for them [**?**]. A broader survey by Zhao et al. reviewed the use of dictionary learning for CT image de-noising and histopathological image classification, and applied low-rank shared dictionary learning to glaucoma diagnosis on fundus images [**?**].

These studies show that dictionary learning methods, and LC-KSVD in particular, remain a relevant classical approach in medical imaging. Their sparse representations offer a natural way to interpret which parts of an image contributed to a decision. This makes dictionary learning a suitable classical baseline for comparison against deep learning-based interpretability methods.

2.7.5 Image Reconstruction

Dictionary learning was first widely applied to image reconstruction problems such as MRI and CT reconstruction. Medical images often contain noise, missing information, or require long acquisition times. Dictionary learning helps overcome these problems by representing image patches as sparse combinations of learned dictionary atoms, allowing high-quality images to be reconstructed from incomplete or degraded data.

Ravishankar and Bresler applied adaptive dictionary learning for compressed sensing MRI reconstruction, where the dictionary was learned directly from the image data instead of using fixed transform bases. Their method achieved improved image quality while reducing MRI acquisition time [45]. Similar sparse representation methods have also been used for low-dose CT reconstruction and image denoising, where learned dictionaries help reduce noise while preserving important anatomical details [47]. These studies show that dictionary learning is useful for reconstruction tasks, especially when interpretability and limited training data are important

2.7.6 Classification and Segmentation

In addition to reconstruction, dictionary learning has been applied to medical image classification and segmentation tasks. In these applications, image patches are converted into sparse representations, and the learned features are used to identify diseases or important anatomical structures. LC-KSVD improved this process by learning both a dictionary and a classifier while ensuring that the sparse codes remain consistent with their class labels [42].

Several studies have shown the effectiveness of dictionary learning for medical image analysis. Tong et al. used discriminative dictionary learning for hippocampus segmentation from brain MRI images, where sparse features captured important anatomical patterns [46].

In CT imaging, Ben-Cohen et al. combined dictionary learning with a fully convolutional network for liver lesion detection, showing that dictionary learning can work together with deep learning methods [48].

Also, dictionary learning has been applied to disease classification tasks such as histopathology analysis and glaucoma diagnosis [47]. These applications demonstrate that dictionary learning provides interpretable features and remains a valuable classical baseline for comparison with modern deep learning models.

2.8 Evaluation Metrics

The evaluation of medical image analysis systems requires multiple metrics because different measures capture different aspects of model behaviour. In chest CT analysis, classification metrics describe the ability of the model to identify abnormal findings, segmentation metrics measure spatial agreement with reference annotations, and interpretability metrics evaluate whether the learned representations provide meaningful explanations.

2.8.1 Classification

For a binary classification problem, let TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively. Accuracy measures the overall proportion of correctly classified samples:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}. \tag{2.4}$$

Although accuracy provides an overall performance estimate, it can become misleading when the dataset contains class imbalance. Therefore, additional metrics based on positive class prediction are considered.

Precision indicates the reliability of positive predictions by measuring the fraction of detected abnormalities that are correct:

$$\text{Precision} = \frac{TP}{TP + FP}. \tag{2.5}$$

Recall, also referred to as sensitivity, measures the ability of the model to recover existing abnormal cases:

$$\text{Recall} = \frac{TP}{TP + FN}. \tag{2.6}$$

The F1-score combines precision and recall into a single measure and provides a balanced evaluation when both false positive and false negative errors are important:

$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{2.7}$$

The Area Under the Receiver Operating Characteristic Curve (AUROC) evaluates the ability of a classifier to separate positive and negative samples over all possible decision thresholds. It is calculated from the relationship between the true positive rate (TPR) and false positive rate (FPR):

$$\text{AUROC} = \int_0^1 \text{TPR}(\text{FPR})\, d\text{FPR}. \tag{2.8}$$

A higher AUROC indicates that the model can better distinguish between abnormal and normal cases without depending on a particular classification threshold.

Average Precision (AP) summarizes the precision-recall relationship and is particularly useful for medical datasets where abnormal cases may represent a smaller proportion of the data:

$$\text{AP} = \sum_n (R_n - R_{n-1}) P_n, \tag{2.9}$$

where P_n and R_n represent precision and recall values at different classification thresholds. Higher average precision indicates better detection of positive cases under class imbalance.

The reliability of predicted probabilities is measured using Expected Calibration Error (ECE). The metric compares predicted confidence with the actual accuracy within confidence intervals:

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} |\mathrm{acc}(B_m) - \mathrm{conf}(B_m)|. \tag{2.10}$$

Lower ECE values indicate that the predicted confidence scores better reflect the actual likelihood of correctness.

2.8.2 Localization and Segmentation

For segmentation evaluation, let P represent the predicted region and G represent the ground-truth annotation. The Dice coefficient measures the overlap between the two regions:

$$\mathrm{Dice}(P, G) = \frac{2|P \cap G|}{|P| + |G|}. \tag{2.11}$$

Intersection over Union (IoU), also known as the Jaccard index, measures the ratio between the intersection and union of the predicted and reference regions:

$$\mathrm{IoU}(P, G) = \frac{|P \cap G|}{|P \cup G|}. \tag{2.12}$$

Dice and IoU provide complementary information about segmentation quality, where higher values indicate better spatial agreement between predicted and reference regions.

2.8.3 Interpretability of Sparse Dictionary Learning

In sparse dictionary learning, interpretability is obtained from the learned dictionary atoms and their corresponding sparse representations. Therefore, evaluation focuses on whether the dictionary provides compact, meaningful, and discriminative representations of CT image patterns.

The reconstruction error measures how effectively the learned dictionary represents the input data. Given an input patch x, dictionary D, and sparse coefficient vector α, the reconstruction error is defined as:

$$\text{Reconstruction Error} = \|x - D\alpha\|_2^2. \tag{2.13}$$

A lower reconstruction error indicates that the dictionary captures important structures present in the image data.

The sparsity of the representation describes how many atoms contribute to the reconstruction. It is measured using the number of non-zero coefficients:

$$\text{Sparsity} = \|\alpha\|_0. \tag{2.14}$$

Lower values indicate that only a small number of atoms are required to explain a sample, making the representation easier to interpret.

Dictionary coherence evaluates the similarity between different atoms in the learned dictionary. The mutual coherence is defined as:

$$\mu(D) = \max_{i \neq j} \frac{|d_i^T d_j|}{\|d_i\|_2 \|d_j\|_2}. \tag{2.15}$$

A lower coherence value indicates that the dictionary contains less redundant atoms and that individual atoms capture more distinct image characteristics.

Atom utilization measures how frequently each dictionary atom contributes to the representation of the dataset:

$$U_k = \sum_{i=1}^{N} \nVdash(\alpha_{k,i} \neq 0), \tag{2.16}$$

where U_k represents the activation frequency of atom k. This metric helps identify whether the learned dictionary is effectively utilized or dominated by a limited number of atoms.

For supervised dictionary learning approaches such as LC-KSVD, class-specific atom activation provides additional interpretability. Disease-related atoms are expected to activate more frequently for samples belonging to their associated class compared with unrelated classes. This indicates that the learned atoms capture discriminative patterns rather than only general image structures.

Finally, visual inspection of reconstructed dictionary atoms is used as a qualitative interpretability assessment. By examining representative atoms, their relationship with anatomical structures or pathological appearances can be analysed, providing an intuitive understanding of the features learned by the model.

2.9 Research Gaps

Chapter 3

Methodology

3.1 Data Acquisition

The quality and consistency of the input data has a direct bearing on how well the dictionary learning stage can later separate abnormalities from normal lung, so a fair amount of the early project time went into getting this part right before any model code was written. This section walks through the datasets that were chosen, how the raw CT volumes were loaded and explored, the different lung segmentation strategies that were tried (and in one case abandoned), and how the final lung volumes were cut down into the 3D patches that feed the K-SVD and LC-KSVD pipelines.

3.1.1 Datasets

Two publicly available chest CT collections were combined for this project rather than relying on a single source, mainly because no single dataset offered both a large enough pool of normal scans and fine-grained, spatially localized annotations for the specific abnormalities of interest.

The first is **CT-RATE**[20], a large-scale chest CT dataset hosted on Hugging Face (`ibrahimhamamci/CT-RATE`) that provides volume-level multi-label annotations across 18 abnormality categories. The released label files cover 47,149 training volumes and 3,039 validation volumes, giving 50,188 labelled scans in total. Table-level counts confirm that the dataset is reasonably well populated for the abnormalities most relevant to this study – lung nodules appear as a positive finding in 21,382 scans and consolidation in 8,319 – while also containing 5,226 scans with no positive label at all, which is what made it a practical source of "normal" lung examples. Because of the dataset's size, only a subset of roughly 2,465 volumes was actually downloaded locally through a small helper script that queries the CT-RATE repository tree via the Hugging Face Hub API and pulls specific volume files on demand.

The second dataset, **ReXGroundingCT**[49], was brought in specifically because it supplies per-finding segmentation masks rather than just volume-level tags. It is organised as 2,992 training, 50 validation and 100 test studies (3,142 in total), each annotated with one or more of 14 finding categories. For this project, attention was narrowed to three categories that best matched the target abnormalities agreed on with the radiologists: atelectasis/consolidation (category `2b`), ground-glass opacity (`2c`) and pulmonary nodules/masses (`2d`). After filtering the metadata down to studies containing at least one of these categories, 1,418 studies remained (1,354 train, 23 validation, 41 test), split roughly as 659 atelectasis/consolidation, 871 GGO and 908 nodule/mass cases – the two abnormality categories occur together less often than one might expect (correlation of about -0.32), which turned out to be convenient for keeping the classes reasonably separable. A further 347 scans in ReXGroundingCT had no annotated findings at all and were folded in alongside the CT-RATE normals to build a combined pool of negative examples. Each finding mask is stored as a 4D NIfTI volume of shape $[F, H, W, D]$, with one binary channel per annotated finding, which later made it straightforward to collapse multiple findings of the same category into a single lesion mask per scan.

An exploratory data analysis (EDA) was carried out before any model work began, and it shaped a number of decisions described later in this chapter. Beyond simple label counts, the EDA looked at volume dimensions, voxel spacing, and the Hounsfield Unit (HU) intensity distribution inside the lungs for each target category. Raw volumes were consistently 512×512 in-plane, but the number of slices varied widely from scan to scan (values such as 220, 238, 251, 466 and 538 slices were all observed), and voxel spacing was not isotropic either – a typical example measured roughly $0.74\,\text{mm} \times 0.74\,\text{mm}$ in-plane against $1.5\,\text{mm}$ between slices. This kind of inconsistency across scanners and acquisition protocols meant that resampling to a common spacing was unavoidable before patches from different scans could be treated as comparable inputs, which is discussed further in Section 3.3. Per-category HU histograms computed over lung voxels also showed that nodules, consolidation and GGO occupy overlapping but distinguishable intensity ranges, which gave some early confidence that a patch-based, intensity-driven representation was a reasonable direction to pursue.

3.1.2 Target Abnormality Selection

Rather than picking abnormality classes purely on the basis of which labels happened to have the most examples, the target classes were decided together with practising radiologists over a series of consultation sessions. The datasets on their own contain dozens of possible findings, many of which are either too rare to model reliably or of limited diagnostic priority compared to others, therefore clinical input was needed to narrow this down to something both meaningful and tractable.

The radiologists were shown the available label distributions from CT-RATE and the finding categories in ReXGroundingCT and asked which abnormalities would be most valuable to detect automatically, and which were common enough in practice to justify the modelling effort. Two considerations came up repeatedly in these discussions: first, that pulmonary nodules and ground-glass/consolidation patterns are among the most frequently missed or delayed findings on routine chest CT, and second, that both have a well-established radiological basis for early intervention, making them good candidates for a proof-of-concept detection system. Based on this feedback, and cross-checked against how well each candidate class was actually represented in the annotated data, the following were fixed as the target classes for this study:

- 2C : Ground-glass opacity (GGO) / consolidation
- 2d : Lung nodule

This clinical grounding mattered in a second, more practical way as well: it directly influenced the segmentation and patch-extraction requirements described in the rest of this section, since both target abnormalities tend to sit close to the lung boundary or alter the surrounding tissue in ways that are easy for naive segmentation methods to mishandle.

3.1.3 CT Volume Loading

All CT scans in this study were distributed in NIfTI format, which keeps both the 3D voxel array and the spatial metadata (orientation, spacing, origin) bundled together in one file. Volumes were read using `nibabel` and converted into float32 HU arrays, with each scan represented as

$$I(x, y, z) \in \mathbb{R}^{H \times W \times D} \tag{3.1}$$

where H, W and D denote the height, width and depth of the volume. Alongside the intensity array, the voxel spacing recorded in the NIfTI header was retained for every scan, since, as noted above, spacing was not consistent across the dataset and this information is needed later to resample volumes onto a common grid.

Where ground-truth finding masks were available (ReXGroundingCT), the corresponding 4D mask volume was loaded through the same pathway and kept aligned with its source CT volume via a shared volume identifier, therefore that a mask could always be traced back to the exact scan and finding it belongs to.

Loading a scan or a mask, however, was never just a matter of reading the NIfTI file and handing back an array – both paths funnel through a shared loading routine that also decides whether resampling and lung preprocessing are required before the data is considered ready to use. Figure 3.1 summarises this routine. After a volume or mask is read with NiBabel, its voxel spacing is compared against the target spacing of 1.5 mm isotropic; if the two already match, the array is passed through unchanged, but if not, it is resampled using trilinear interpolation for CT intensities or nearest-neighbour interpolation for masks, mirroring the distinction made earlier for `resample_volume` and `resample_mask` – nearest-neighbour is used specifically to avoid inventing intermediate label values between the discrete finding categories in a mask. Only volumes, and never masks, are then routed through the preprocessing block, where lung segmentation, HU clipping to $[-1000, 500]$ and min-max normalisation are applied in sequence before the final volume is returned; masks skip this block entirely and are returned as soon as resampling is done, since a binary or categorical finding mask has no HU values to clip or normalise and does not need to be run through the lung segmentation model. Keeping this branching logic in one place meant that every scan and every mask used later in the pipeline, regardless of which dataset it came from, had been through exactly the same spacing correction and (where relevant) the same preprocessing steps, so patches extracted from different scans remained directly comparable.

3.1.4 Preprocessing

Before any patches could be extracted, the raw volumes had to be reduced to just the lung region. A whole-volume approach was ruled out fairly early: bones, soft tissue and background air all add computational overhead without contributing anything useful to dictionary learning, and worse, they risk letting the model pick up spurious patterns unrelated to lung pathology. Three different approaches to isolating the

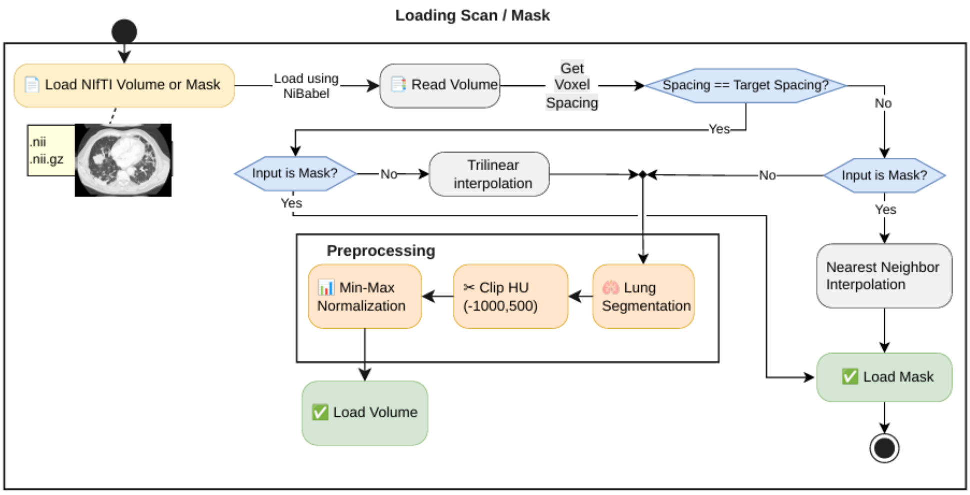

Figure 3.1: Shared loading pipeline for CT volumes and finding masks, showing the spacing check, resampling branch (trilinear for volumes, nearest-neighbour for masks) and the preprocessing block applied only to volumes.

lungs were tried over the course of the project, in the order described below, before settling on a final method.

Classical Image Processing-Based Lung Segmentation

The first attempt used a fairly standard HU-threshold pipeline, since lung tissue is normally well separated from surrounding structures in intensity. Each volume was first thresholded to keep only voxels between -900 and -200 HU, the range typically associated with aerated lung parenchyma:

$$M_{thresh}(x,y,z) = \begin{cases} 1, & -900 \leq I(x,y,z) \leq -200 \\ 0, & \text{otherwise} \end{cases} \tag{3.2}$$

This crude threshold mask is noisy on its own, so it was cleaned up with a short sequence of morphological steps applied slice-by-slice and volume-wise using `scipy.ndimage` and `skimage.segmentation`. Border-connected regions were removed on each axial slice with `clear_border` (mainly to drop the surrounding air and imaging-table artefacts that also fall inside the HU window), after which only the two largest connected components were kept – under the assumption that these correspond to the left and right lung:

A 3D binary closing (a $3\times3\times3$ connectivity structure, 2 iterations) was then used to smooth the boundary and reconnect thin gaps, followed by `binary_fill_holes` to plug any internal cavities left inside the mask – these two steps are exactly what a lung mask needs, since vessels, bronchi and small nodules would otherwise show up as holes in an otherwise solid lung region.

On visibly normal scans, this five-stage pipeline (threshold $\rightarrow$ clear border $\rightarrow$ largest components $\rightarrow$ closing $\rightarrow$ hole filling) produced reasonable-looking lung masks when checked slice-by-slice. The problem showed up once pathology entered the picture. Consolidation and areas of lung opacity raise the local tissue density towards that of soft tissue, so the affected voxels are not registered as lung parenchyma at all under the initial -900 to -200 HU window – they get cut out of the mask at the very first thresholding step, before any of the later stages ever see them.

What happens next depends on where the abnormality sits and how large it is. When the affected region is small and fully surrounded by normal, correctly thresholded lung tissue, it behaves like just another gap in the mask, and the binary closing and hole-filling steps are usually enough to patch it back in – the same mechanism that is meant to fill in small vessels and airways ends up rescuing these interior lesions almost by accident. But when the abnormality is larger, or sits close to the lung boundary rather than fully enclosed within it, closing and hole-filling can no longer bridge the gap: the affected region gets left connected to the outside of the lung rather than treated as an enclosed hole, so it is excluded from the final mask along with everything else outside the lung boundary. This is exactly the pattern that shows up when the intermediate masks are stepped through and overlaid on the source slice – a large opacity running along the pleural surface is visibly missing from the final lung segmentation and from the CT after the mask is applied, while the ground-truth annotation for that same region sits entirely outside the segmented lung. The five-stage pipeline was checked this way across several scans, and the failure was consistent whenever the abnormal region was either large or adjacent to the lung border, rather than an isolated case.

For a project whose entire purpose is to detect exactly these kinds of abnormalities, a segmentation step that only reliably preserves lesions small enough to be treated as incidental holes is not something that could be fixed with more morphological post-processing – the information is already lost at the thresholding step, since abnormal tissue often does not fall inside the normal lung parenchyma HU window in the first place. This is what motivated the move away from purely intensity-based segmentation and towards a learned segmentation model, LungMask's `LMInferer`, described next.

LungMask-Based Lung Segmentation

The method that was ultimately adopted is the `LungMask` library, accessed through its `LMInferer` interface. Unlike the classical pipeline, LungMask is a trained segmentation network built specifically for whole-lung extraction, and it proved noticeably more stable when abnormal tissue was present – consolidation and GGO regions inside the lung boundary were retained rather than excluded, which directly addressed the main failure observed with classical image processing approach.

Given a CT volume, LungMask returns a label map with three values: 0 for background, 1 for the left lung and 2 for the right lung. This was converted into a binary mask and applied to the original volume:

$$I_{lung} = I \times M_{lung} \tag{3.3}$$

where I is the original CT volume and M_{lung} is the binary lung mask produced by LungMask. Voxels falling outside the mask were reset to a fixed background value of -1000 HU (approximating air), after which the volume was clipped to the range $[-1000, 500]$ HU and linearly rescaled to $[0, 1]$. This combination of segmentation, HU clipping and normalisation was wrapped into a single preprocessing routine that also resamples each volume to an isotropic 1.5 mm spacing beforehand, therefore that every scan reaching the patch extraction stage has a consistent voxel grid regardless of its original acquisition parameters.

3.1.5 Patch Extraction

With a clean, normalised lung volume in hand, the final data acquisition step was to break each volume down into smaller 3D patches. Feeding entire volumes directly into the dictionary learning stage was never really practical – a single scan can contain tens of millions of voxels – and patch-based sampling has the added benefit of letting the model focus on local structural patterns rather than whole-organ context, which fits naturally with how K-SVD and LC-KSVD represent signals as sparse combinations of small dictionary atoms.

Each patch is a fixed $16 \times 16 \times 16$ voxel cube, which at the resampled 1.5 mm spacing corresponds to roughly a $24\,\text{mm}^3$ region of lung tissue – large enough to capture a typical small nodule or a local patch of opacity, without dragging in unrelated anatomy. Patches were sampled differently depending on whether they were meant to represent normal tissue or a target abnormality:

1. **Normal patches** were drawn on a non-overlapping grid across the entire lung volume, with a stride equal to the patch size. A patch was discarded if more than half of its voxels were near-zero after masking, since that generally indicates the patch sits mostly outside the lung region rather than within actual tissue.

2. **Abnormal patches** were sampled only from scans carrying a ground-truth finding mask. Since a single scan can have several individual findings that belong to the same target category, the relevant finding channels of the 4D mask were first OR-combined into one binary lesion mask per category present in that scan. A tight axis-aligned bounding box was then computed around the foreground of that lesion mask, and the patch window was slid across the box with a stride of 4 voxels in each direction – markedly denser than the normal-patch grid, since the region of interest here is already small and every overlapping view of it is potentially useful. Each candidate patch was only kept if the fraction of lesion-labelled voxels within its corresponding mask region reached a category-specific threshold: 50% for ground-glass opacity/consolidation patches, but only 10% for nodule/mass patches, reflecting how much smaller and more localised a pulmonary nodule typically is compared with a spread-out opacity – a fixed 50% cutoff would have discarded almost every genuine nodule patch. A scan positive for both target categories simply contributes patches to both, each filtered against its own bounding box and threshold.

Figure 3.2 lays out this branching more concretely. The first decision is simply whether the loaded scan is one of the designated normal volumes or one that carries a ground-truth abnormality mask. Normal scans go straight into a regular sampling grid built over the whole volume, whereas scans with an annotation instead have a 3D bounding box generated directly around the ground-truth lesion, so that patch sampling is concentrated on the region that actually contains the finding rather than scattered across the entire lung. Whichever route a patch takes, it still has to pass one final check before it is accepted: patches coming from the normal grid are rejected if more than half of their voxels sit at background level ($-1000\,\text{HU}$, i.e. outside the lung after masking), while patches coming from a lesion bounding box are rejected unless their lesion-voxel fraction meets that category's threshold. Only patches that clear this check are appended to the patch matrix; everything else is discarded before it ever reaches the dictionary learning stage. Structuring the two branches this way keeps the normal and abnormal sampling logic independent while funnelling both into the same final acceptance criterion and the same output matrix.

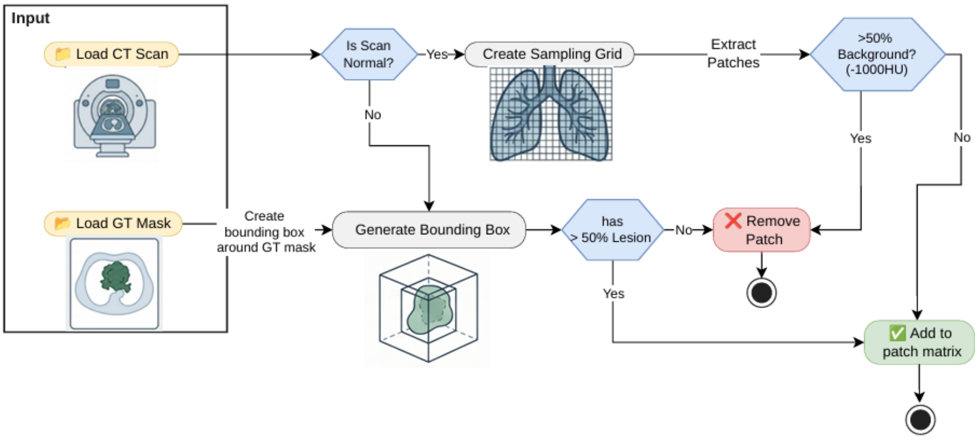

Figure 3.2: Patch extraction workflow, showing the normal-scan sampling grid and the lesion bounding-box branch for annotated scans, each filtered by its own background/lesion-fraction threshold before patches are added to the patch matrix.

Formally, each extracted patch can be written as

$$P_i \in \mathbb{R}^{p \times p \times p}, \quad p = 16 \tag{3.4}$$

and is subsequently flattened into a vector representation before being handed to the dictionary learning algorithms:

$$x_i \in \mathbb{R}^n, \quad n = p^3 = 4096 \tag{3.5}$$

Patches extracted this way were saved per scan and per class (normal, GGO, nodule) as compressed archives containing the flattened feature matrix, class labels, the originating scan identifiers and the voxel coordinates of each patch, and were later concatenated across classes to build the full training matrix used for K-SVD, frozen dictionary learning, and LC-KSVD. This patch-based representation keeps memory requirements manageable while still allowing the learned dictionary atoms to specialise in the local structural signatures associated with each target pulmonary abnormality.

¡¡¡¡¡¡¡ HEAD

3.2 Classification

3.2.1 Sparse Coding

3.2.2 Training

LinearSVC

Logistic Regression

3.3 Inference

3.3.1 Classification on Test Set

3.3.2 Statistical Analysis

3.4 Deep Learning Models

3.4.1 nnU-Net

3.4.2 Merlin

3.4.3 BiomedParse

3.4.4 CT-CLIP

3.4.5 Med-SAM

3.5 Evaluation Framework

3.5.1 Metrics

Classification Metrics

Segmentation Metrics

Interpretability Metrics

3.5.2 Pipeline

Adaptor

Runner

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3.6 Evaluation Metrics

The evaluation of medical image analysis systems requires multiple metrics because different measures capture different aspects of model behaviour. In chest CT analysis, classification metrics describe the ability of the model to identify abnormal findings, segmentation metrics measure spatial agreement with reference annotations, and interpretability metrics evaluate whether the learned representations provide meaningful explanations.

3.6.1 Classification

For a binary classification problem, let TP, TN, FP, and FN represent true positives, true negatives, false positives, and false negatives, respectively. Accuracy measures the overall proportion of correctly classified samples:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}. \tag{3.6}$$

Although accuracy provides an overall performance estimate, it can become misleading when the dataset contains class imbalance. Therefore, additional metrics based on positive class prediction are considered.

Precision indicates the reliability of positive predictions by measuring the fraction of detected abnormalities that are correct:

$$\text{Precision} = \frac{TP}{TP + FP}. \tag{3.7}$$

Recall, also referred to as sensitivity, measures the ability of the model to recover existing abnormal cases:

$$\text{Recall} = \frac{TP}{TP + FN}. \tag{3.8}$$

The F1-score combines precision and recall into a single measure and provides a balanced evaluation when both false positive and false negative errors are important:

$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}. \tag{3.9}$$

The Area Under the Receiver Operating Characteristic Curve (AUROC) evaluates the ability of a classifier to separate positive and negative samples over all possible decision thresholds. It is calculated from the relationship between the true positive rate (TPR) and false positive rate (FPR):

$$\text{AUROC} = \int_0^1 \text{TPR}(\text{FPR})\, d\text{FPR}. \tag{3.10}$$

A higher AUROC indicates that the model can better distinguish between abnormal and normal cases without depending on a particular classification threshold.

Average Precision (AP) summarizes the precision-recall relationship and is particularly useful for medical datasets where abnormal cases may represent a smaller proportion of the data:

$$\text{AP} = \sum_n (R_n - R_{n-1}) P_n, \tag{3.11}$$

where P_n and R_n represent precision and recall values at different classification thresholds. Higher average precision indicates better detection of positive cases under class imbalance.

The reliability of predicted probabilities is measured using Expected Calibration Error (ECE). The metric compares predicted confidence with the actual accuracy within confidence intervals:

$$\mathrm{ECE} = \sum_{m=1}^{M} \frac{|B_m|}{N} |\mathrm{acc}(B_m) - \mathrm{conf}(B_m)|. \tag{3.12}$$

Lower ECE values indicate that the predicted confidence scores better reflect the actual likelihood of correctness.

3.6.2 Localization and Segmentation

For segmentation evaluation, let P represent the predicted region and G represent the ground-truth annotation. The Dice coefficient measures the overlap between the two regions:

$$\mathrm{Dice}(P, G) = \frac{2|P \cap G|}{|P| + |G|}. \tag{3.13}$$

Intersection over Union (IoU), also known as the Jaccard index, measures the ratio between the intersection and union of the predicted and reference regions:

$$\mathrm{IoU}(P, G) = \frac{|P \cap G|}{|P \cup G|}. \tag{3.14}$$

Dice and IoU provide complementary information about segmentation quality, where higher values indicate better spatial agreement between predicted and reference regions.

3.6.3 Interpretability of Sparse Dictionary Learning

In sparse dictionary learning, interpretability is obtained from the learned dictionary atoms and their corresponding sparse representations. Therefore, evaluation focuses on whether the dictionary provides compact, meaningful, and discriminative representations of CT image patterns.

The reconstruction error measures how effectively the learned dictionary represents the input data. Given an input patch x, dictionary D, and sparse coefficient vector α, the reconstruction error is defined as:

$$\text{Reconstruction Error} = \|x - D\alpha\|_2^2. \tag{3.15}$$

A lower reconstruction error indicates that the dictionary captures important structures present in the image data.

The sparsity of the representation describes how many atoms contribute to the reconstruction. It is measured using the number of non-zero coefficients:

$$\text{Sparsity} = \|\alpha\|_0. \tag{3.16}$$

Lower values indicate that only a small number of atoms are required to explain a sample, making the representation easier to interpret.

Dictionary coherence evaluates the similarity between different atoms in the learned dictionary. The mutual coherence is defined as:

$$\mu(D) = \max_{i \neq j} \frac{|d_i^T d_j|}{\|d_i\|_2 \|d_j\|_2}. \tag{3.17}$$

A lower coherence value indicates that the dictionary contains less redundant atoms and that individual atoms capture more distinct image characteristics.

Atom utilization measures how frequently each dictionary atom contributes to the representation of the dataset:

$$U_k = \sum_{i=1}^{N} \nVdash(\alpha_{k,i} \neq 0), \tag{3.18}$$

where U_k represents the activation frequency of atom k. This metric helps identify whether the learned dictionary is effectively utilized or dominated by a limited number of atoms.

For supervised dictionary learning approaches such as LC-KSVD, class-specific atom activation provides additional interpretability. Disease-related atoms are expected to activate more frequently for samples belonging to their associated class compared with unrelated classes. This indicates that the learned atoms capture discriminative patterns rather than only general image structures.

Finally, visual inspection of reconstructed dictionary atoms is used as a qualitative interpretability assessment. By examining representative atoms, their relationship with anatomical structures or pathological appearances can be analysed, providing an intuitive understanding of the features learned by the model. ¿¿¿¿¿¿¿ b058374df99af8ff6896c419bd5551bef9d32b98

Chapter 4

Conclusion

Appendix A

Essential English Grammar - A Handout

The Appendices are numbered as A, B,.. and so on. Sections and Subsections of the appendix then would be A.1, A.1.1, A.2 etc.

A.1 Simple Present Tense

A.2 Perfect Tense

A.3 Past Past Perfect Tense

A.4 Passive Voice

A.5 Technical Writing Fundamentals

A.6 Proof Reading

Appendix B

Graphical Interpretation of Data

The graphical interpretation given below shows the relationship between x and y is acceptable to the purpose of the project

B.1 Data Analysis

B.1.1 Statistical Analysis

B.1.2 Empirical Analysis

B.2 The X vs Y vs Z

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